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Sheet storage device and image forming apparatus including same

Abstract

A sheet storage device includes a lift plate, a push-up member, a first gear, a second gear which meshes with the first gear, a transmission mechanism which includes a third gear that meshes with the second gear and a turning restriction member. The first gear includes a support portion. The turning restriction member includes: an annular portion which is slidable on the outer circumferential surface of the support portion in a direction of rotation; and a protrusion portion which protrudes outward in a radial direction from the annular portion to be able to engage with the second gear. The protrusion portion has a shape such that the protrusion portion cannot pass through a gap between the second gear and the annular portion and is located, relative to the gap, on a downstream side in a direction of rotation of the second gear caused by the power of a motor.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-105983 (filed on Jun. 28, 2023), the entire contents of which are incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a sheet storage device and an image forming apparatus which includes such a sheet storage device.

(3) A conventional sheet storage device includes a lift plate which can be raised and lowered. Sheets are placed on the lift plate. The lift plate is raised to hold the height position of the uppermost sheet on the lift plate in a predetermined position.

(4) The conventional sheet storage device includes a paper feed roller which makes contact with the sheet in the predetermined position from above. The paper feed roller is rotated in a state where the paper feed roller is in contact with the sheet to feed the sheet. When the uppermost sheet on the lift plate is not in the predetermined position, the paper feed roller idles so as not to feed the sheet.

SUMMARY

(5) A sheet storage device according to a first aspect of the present disclosure includes a lift plate, a push-up member, a first gear, a second gear, a transmission mechanism and a turning restriction member. A sheet is placed on the lift plate. The push-up member is turnably supported below the lift plate, receives transmission of the power of a motor and is turned in a lift-up direction to push up the lift plate. The first gear is rotated in one direction by the power of the motor. The second

gear meshes with the first gear. The transmission mechanism includes at least a third gear which meshes with the second gear to transmit the power of the motor to the push-up member. The turning restriction member restricts turning of the push-up member in a direction opposite to the lift-up direction. The first gear includes a cylindrical support portion with the center axis of the first gear being a cylindrical axis. The turning restriction member includes an annular portion and a protrusion portion. The support portion is fitted to the annular portion, and the annular portion is slidable on the outer circumferential surface of the support portion in the direction of the rotation. The protrusion portion protrudes outward in a radial direction from the annular portion to be able to engage with the second gear. The protrusion portion has a shape such that the protrusion portion cannot pass through a gap between the second gear and the annular portion in a direction orthogonal to an axial direction, and is located, relative to the gap, on a downstream side in a direction of rotation of the second gear caused by the power of the motor.

(6) An image forming apparatus according to a second aspect of the present disclosure includes the sheet storage device described above, and conveys the sheet and prints an image on the sheet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic view of an image forming apparatus which includes a sheet storage device according to an embodiment;
- (2) FIG. 2 is a plan view when a gear train coupled to a push-up member in the sheet storage device according to the embodiment is viewed in an axial direction;
- (3) FIG. 3 is a perspective view of a first gear and an area therearound in the sheet storage device according to the embodiment;
- (4) FIG. 4 is a cross-sectional perspective view of the first gear and the area therearound in the sheet storage device according to the embodiment;
- (5) FIG. 5 is a perspective view of a turning restriction member in the sheet storage device according to the embodiment;
- (6) FIG. 6 is a perspective view of the turning restriction member and an area therearound in the sheet storage device according to the embodiment;
- (7) FIG. 7 is a diagram showing a state where a switching mechanism in the sheet storage device according to the embodiment transmits the power of a motor; and
- (8) FIG. 8 is a diagram showing a state where the switching mechanism in the sheet storage device according to the embodiment blocks the transmission of the power of the motor.

DETAILED DESCRIPTION

(9) A sheet storage device **100** according to an embodiment of the present disclosure and an image forming apparatus **1000** which includes the sheet storage device **100** will be described below with reference to FIGS. 1 to 8. The image forming apparatus **1000** is installed on a floor surface. A direction perpendicular to the floor surface (flat surface) on which the image forming apparatus **1000** is installed corresponds to an up/down direction.

(10) <Configuration of Image Forming Apparatus>

(11) The image forming apparatus **1000** of the present embodiment is a printer. The image forming apparatus **1000** can perform a print job. The image forming apparatus **1000** performs the print job to print an image on a sheet S. The printing system of the image forming apparatus **1000** is, for example, an electrophotographic system. The printing system of the image forming apparatus **1000** may be an inkjet system.

(12) The image forming apparatus **1000** includes the sheet storage device **100**. The sheet storage device **100** stores the sheet S which is used in the print job. For example, the sheet S is a sheet (that is, made of paper). The image forming apparatus **1000** conveys the sheet S stored in the sheet

storage device **100** along a conveyance path, and prints an image on the sheet S being conveyed. In FIG. **1**, the conveyance path for the sheet S is schematically indicated by a dotted line with an arrow. A direction indicated by the arrow is a sheet conveyance direction.

(13) Hereinafter, a direction which is orthogonal to the sheet conveyance direction and is also orthogonal to the up/down direction may be referred to as a width direction. In FIG. **1**, a direction perpendicular to the plane of the sheet is the width direction. For example, FIG. **1** corresponds to a diagram when the image forming apparatus **1000** is viewed from the front. In other words, the width direction is also a forward/backward direction.

(14) The image forming apparatus **1000** includes a paper feed roller **1001**. The paper feed roller **1001** is supported to be rotatable around an axis extending in the width direction. The paper feed roller **1001** feeds the sheet S stored in the sheet storage device **100** to the conveyance path. The paper feed roller **1001** makes contacts with the sheet S stored in the sheet storage device **100** from above, and is rotated. In this way, the sheet S is fed from the sheet storage device **100** to the conveyance path.

(15) The image forming apparatus **1000** includes a registration roller pair **1002**. The registration roller pair **1002** is supported to be rotatable around an axis extending in the width direction. The registration roller pair **1002** includes a pair of rollers which are pressed against each other. The registration roller pair **1002** nips the sheet S fed from the paper feed roller **1001**, and is rotated. In this way, the registration roller pair **1002** conveys the sheet S. One roller of the registration roller pair **1002** is a drive roller to which power is transmitted from a motor M and which is rotated, and the other is a driven roller which is rotated by following the one registration roller.

(16) The image forming apparatus **1000** includes an image formation unit **1003**. The image formation unit **1003** includes a photosensitive drum **1003a** and a transfer roller **1003b**. Each of the photosensitive drum **1003a** and the transfer roller **1003b** is supported to be rotatable around an axis extending in the width direction. The photosensitive drum **1003a** and the transfer roller **1003b** are pressed against each other. The photosensitive drum **1003a** and the transfer roller **1003b** nip the sheet S fed from the registration roller pair **1002**, and are rotated. In this way, the image formation unit **1003** conveys the sheet S.

(17) The image formation unit **1003** further includes a charging device, an exposure device and a development device though they are not shown in the figure. The charging device charges the circumferential surface of the photosensitive drum **1003a**. The exposure device forms an electrostatic latent image on the circumferential surface of the photosensitive drum **1003a**. The development device develops the electrostatic latent image on the circumferential surface of the photosensitive drum **1003a** into a toner image. The photosensitive drum **1003a** nips the sheet S between the transfer roller **1003b** and itself, is rotated together with the transfer roller **1003b** and thereby transfers the toner image to the sheet S.

(18) The image forming apparatus **1000** includes a fixing roller pair **1004**. The fixing roller pair **1004** is supported to be rotatable around an axis extending in the width direction. The fixing roller pair **1004** includes a heating roller and a pressure roller. The heating roller incorporates a heater. The pressure roller is pressed against the heating roller, and the fixing roller pair **1004** nips the sheet S fed from the image formation unit **1003** and is rotated. In other words, the fixing roller pair **1004** heats and pressurizes the sheet S. In this way, the fixing roller pair **1004** fixes, to the sheet S, the toner image transferred to the sheet S. Thereafter, the sheet S is ejected to an ejection tray ET.

(19) <Configuration of Sheet Storage Device>

(20) The sheet storage device **100** includes a sheet cassette CA. The sheet cassette CA stores the sheet S. The sheet cassette CA is removable relative to the main body of the image forming apparatus **1000**. When the print job is performed, and thus no sheet S is in the sheet cassette CA, the sheet cassette CA is pulled out from the main body of the apparatus, the sheets S are stored in the sheet cassette CA and thereafter, the sheet cassette CA is returned into the main body of the apparatus.

(21) The sheet storage device **100** includes a lift plate **101**. The lift plate **101** is arranged in the sheet cassette CA. The sheet S is stored in the sheet cassette CA, and thus the sheet S is placed on the lift plate **101**. The lift plate **101** is turnable around an axis extending in the width direction. The lift plate **101** is turned with an end portion on an upstream side in a sheet feeding direction used as a support point so as to vertically swing an end portion on a downstream side in the sheet feeding direction. In FIG. 1, a direction which extends from the left side of the figure toward the right side is the sheet feeding direction.

(22) While the print job is being performed, the turning of the lift plate **101** in a direction in which the end portion on the downstream side in the sheet feeding direction is raised and the stop of the turning are repeated. In this way, the lift plate **101** maintains the contact of the paper feed roller **1001** with the uppermost sheet S on the lift plate **101**. Even when the number of sheets S on the lift plate **101** is decreased, the lift plate **101** is raised, and thus a state where the sheet S can be fed from the sheet cassette CA to the conveyance path is maintained.

(23) The sheet storage device **100** includes a push-up member **102**. The push-up member **102** is arranged in the sheet cassette CA. The push-up member **102** is supported below the lift plate **101** so as to be turnable around an axis extending in the width direction. The push-up member **102** is turned with an end portion on the upstream side in the sheet feeding direction used as a support point so as to vertically swing an end portion on the downstream side in the sheet feeding direction. The push-up member **102** is turned in a lift-up direction to push up the lift plate **101**. The push-up member **102** may be turned with the end portion on the downstream side in the sheet feeding direction used as the support point so as to vertically swing the end portion on the upstream side in the sheet feeding direction. Regardless of the direction of turning of the push-up member **102**, the lift plate **101** can be pushed up by the push-up member **102**.

(24) The sheet storage device **100** receives the power of the motor M which rotates the registration roller pair **1002**. The power of the motor M is transmitted to the push-up member **102**, and thus the push-up member **102** is turned in the lift-up direction. However, the present disclosure is not limited to this configuration. The power of a dedicated lift-up motor (not shown) may be transmitted to the push-up member **102**, and thus the push-up member **102** may be turned in the lift-up direction.

(25) <Turning of Push-Up Member in Lift-Up Direction>

(26) The sheet storage device **100** includes a gear train which transmits the power of the motor M to the turning shaft **102a** of the push-up member **102**. The gear train of the sheet storage device **100** will be described below with reference to FIGS. 2 to 4. Each of all the gears of the sheet storage device **100** is rotatable around an axis extending in the width direction. In other words, in all the gears of the sheet storage device **100**, the width direction is an axial direction.

(27) In the following description, the direction of rotation of each of the gears of the sheet storage device **100** is based on a case where the rotation is viewed from one side in the axial direction. In the following description, regarding the direction of rotation of each of the gears of the sheet storage device **100**, a counterclockwise direction may be simply referred to as one direction, and a clockwise direction may be simply referred to as the other direction.

(28) The sheet storage device **100** includes a first gear **1** made of resin. Specifically the sheet storage device **100** includes a cylindrical member **10** made of resin which integrally includes the first gear **1**. The cylindrical member **10** (that is, the first gear **1**) is supported to be rotatable around the axis of a shaft **10A** extending in the width direction. The cylindrical member **10** is a stepped cylindrical member. The cylindrical member **10** includes a small diameter portion and a large diameter portion which has an outside diameter larger than the small diameter portion. The first gear **1** is provided on the outer circumferential surface of the small diameter portion in the cylindrical member **10**.

(29) The cylindrical member **10** includes the large diameter portion on the one side in the axial direction (that is, a cylindrical axis direction) relative to the small diameter portion in which the

first gear **1** is provided. In other words, the first gear **1** includes, on the one side in the axial direction, the cylindrical portion (that is, the large diameter portion) which has an outside diameter larger than the first gear **1**. Although details will be described later, the cylindrical portion on the one side in the axial direction included in the first gear **1** is a support portion **11**.

(30) The power of the motor **M** is transmitted to the first gear **1**, and thus the first gear **1** is rotated in one direction. The first gear **1** is rotated in the one direction, and thus the push-up member **102** is turned in the lift-up direction.

(31) The sheet storage device **100** includes a second gear **2** made of resin. The second gear **2** meshes with the first gear **1**. For example, the second gear **2** has its rotational shaft below the rotational shaft of the first gear **1** (that is, the shaft **10A**). The first gear **1** which has received the power of the motor **M** is rotated in the one direction, and thus the second gear **2** is rotated in the other direction.

(32) The second gear **2** is a stepped gear. The second gear **2** is a two-stage gear. The second gear **2** includes, as an input side gear **21**, a part which meshes with the first gear **1**. The second gear **2** includes, as an output side gear **22**, a part which meshes with a third gear **3**. A cylindrical boss portion protrudes from the input side gear **21** toward the one side in the axial direction. The output side gear **22** is provided on the outer circumferential surface of the boss portion. In other words, the outside diameter of the input side gear **21** is larger than that of the output side gear **22**.

(33) The sheet storage device **100** includes a transmission mechanism (the symbol of which is omitted) which includes a plurality of gears. The transmission mechanism transmits the power of the motor **M** to the turning shaft **102a** of the push-up member **102** to turn the push-up member **102** in the lift-up direction. The transmission mechanism includes at least the third gear **3** made of resin which meshes with the second gear **2**. The third gear **3** meshes with the output side gear **22**. The second gear **2** is rotated in the other direction, and thus the third gear **3** is rotated in the one direction.

(34) The transmission mechanism individually includes a fourth gear **4** and a fifth gear **5** which are made of resin. The fourth gear **4** meshes with the third gear **3**. The third gear **3** is rotated in the one direction, and thus the fourth gear **4** is rotated in the other direction. The fifth gear **5** meshes with the fourth gear **4**. The fourth gear **4** is rotated in the other direction, and thus the fifth gear **5** is rotated in the one direction. The number of gears in the transmission mechanism is not particularly limited.

(35) The turning shaft **102a** of the push-up member **102** is coupled to the fifth gear **5** via an unillustrated gear made of resin. In this way, the power of the motor **M** is transmitted to the turning shaft **102a** of the push-up member **102**. The power of the motor **M** is transmitted to the push-up member **102**, and thus the push-up member **102** is turned in the lift-up direction to push up the lift plate **101**.

(36) The sheet storage device **100** includes a switching mechanism **7**. The switching mechanism **7** switches between the transmission of the power from the motor **M** to the first gear **1** and the blocking of the transmission. The power is transmitted from the motor **M** to the first gear **1**, and thus the first gear **1** is rotated in the one direction. When the transmission of the power from the motor **M** to the first gear **1** is blocked, the first gear **1** is not rotated in the one direction. When the first gear **1** is not rotated in the one direction, the push-up member **102** is not turned in the lift-up direction, and thus the lift plate **101** is not raised.

(37) The configuration of the switching mechanism **7** is not particularly limited. It is sufficient that it is possible to switch between the transmission of the power from the motor **M** to the first gear **1** and the blocking of the transmission. For example, the switching mechanism **7** includes a planetary gear mechanism. The configuration of the switching mechanism **7** will be described below. The configuration of the switching mechanism **7** which will be described below is an example. As the switching mechanism **7**, a mechanism which includes an electromagnetic clutch can also be used.

(38) The switching mechanism **7** includes the planetary gear mechanism which is coupled to the

first gear **1**. Specifically, the switching mechanism **7** includes an internal gear **71**. The internal gear **71** is supported to be rotatable around the axis of the shaft **10A**. The internal gear **71** is provided on the inner circumferential surface of the large diameter portion in the cylindrical member **10** which includes the first gear **1**. In other words, the cylindrical member **10** integrally includes the internal gear **71**. Since the internal gear **71** is formed of the same material as the cylindrical member **10**, the internal gear **71** is made of resin.

(39) The switching mechanism **7** includes a sun gear **72** made of resin. The sun gear **72** is supported to be rotatable around the axis of the shaft **10A**. The sun gear **72** is arranged inside the internal gear **71** (that is, inside the large diameter portion of the cylindrical member **10**).

(40) The switching mechanism **7** includes a plurality of planetary gears **73** which are made of resin. Inside the internal gear **71** (that is, inside the large diameter portion of the cylindrical member **10**), the planetary gears **73** are arranged to surround the sun gear **72**. The planetary gears **73** mesh with the sun gear **72**, and also mesh with the internal gear **71**.

(41) The switching mechanism **7** includes a planetary carrier **74** made of resin. The planetary carrier **74** is supported to be rotatable around the axis of the shaft **10A**. The planetary carrier **74** rotatably supports the planetary gears **73**. In this way, the planetary gears **73** can revolve while rotating around the sun gear **72**. The planetary gears **73** revolve around the sun gear **72**, and thus the planetary carrier **74** is rotated.

(42) The planetary carrier **74** integrally includes a power input gear **730**. The power input gear **730** is supported to be rotatable around the axis of the shaft **10A**. The power input gear **730** is arranged on the one side in the axial direction relative to the cylindrical member **10** (the large diameter portion thereof). Although a part of the planetary carrier **74** which supports the planetary gears **73** is arranged inside the cylindrical member **10**, a part of the planetary carrier **74** which includes the power input gear **730** is arranged outside the cylindrical member **10**. Since the power input gear **730** is formed of the same material as the planetary carrier **74**, the power input gear **730** is made of resin.

(43) The power input gear **730** meshes with a drive gear **700** (see FIG. 2). The drive gear **700** is a gear which is attached to the rotational shaft of the drive roller in the registration roller pair **1002**, and is rotated together with the drive roller. In other words, the drive gear **700** is a gear on the most upstream side in a direction in which the power is transmitted from the motor **M** to the turning shaft **102a** of the push-up member **102**.

(44) Here, the sun gear **72** integrally includes an engagement rotation member **720** which is rotatable around the axis of the shaft **10A**. The engagement rotation member **720** is rotated together with the sun gear **72**. Since the sun gear **72** is integral with the engagement rotation member **720**, when the engagement rotation member **720** is in a state where the engagement rotation member **720** cannot be rotated, the sun gear **72** is not rotated.

(45) The engagement rotation member **720** is arranged on the one side in the axial direction relative to the power input gear **730**. The engagement rotation member **720** includes a boss portion which extends to the inside of the large diameter portion of the cylindrical member **10**. The sun gear **72** is provided on the outer circumferential surface of the boss portion extending from the engagement rotation member **720**. In this way, the sun gear **72** is arranged inside the large diameter portion of the cylindrical member **10** to mesh with the planetary gears **73**. Since the engagement rotation member **720** is formed of the same material as the sun gear **72**, the engagement rotation member **720** is made of resin.

(46) The engagement rotation member **720** includes, on its outer circumferential surface, a plurality of protrusions **721** which protrude outward in its radial direction. The protrusions **721** are spaced in the direction of rotation of the engagement rotation member **720**. In other words, the engagement rotation member **720** includes protrusions and recesses in the outer circumferential surface. Furthermore, in other words, the outer circumferential surface of the engagement rotation member **720** can engage with a claw member **75** which will be described later.

(47) While the power input gear **730** is being rotated, the engagement rotation member **720** is brought into the state where the engagement rotation member **720** cannot be rotated (that is, a state where the sun gear **72** cannot be rotated), and thus the first gear **1** is rotated in the one direction. In other words, the power of the motor **M** is transmitted to the first gear **1**, and thus the first gear **1** is rotated in the one direction. In this way, the second gear **2** is rotated in the other direction, the third gear **3** is rotated in the one direction and thus the push-up member **102** is turned in the lift-up direction. Consequently, the lift plate **101** is raised.

(48) While the power input gear **730** is being rotated, the engagement rotation member **720** is brought into a state where the engagement rotation member **720** can be rotated (that is, a state where the sun gear **72** can be rotated), and thus the power of the motor **M** is not transmitted to the first gear **1**. In other words, the transmission of the power from the motor **M** to the first gear **1** is blocked. In this way, the power for turning the push-up member **102** in the lift-up direction is not transmitted to the turning shaft **102a** of the push-up member **102**. Here, the rotation of the first gear **1** is not locked, and thus the first gear **1** is brought into a state where the first gear **1** can be rotated in the other direction opposite to the one direction.

(49) In this configuration, when the lift plate **101** is desired to be raised (that is, when the push-up member **102** is desired to be turned in the lift-up direction), the engagement rotation member **720** is preferably brought into the state where the engagement rotation member **720** cannot be rotated. When the raising of the lift plate **101** is desired to be stopped (that is, when the turning of the push-up member **102** in the lift-up direction is desired to be stopped), the engagement rotation member **720** is preferably brought into the state where the engagement rotation member **720** can be rotated.

(50) Hence, the switching mechanism **7** includes the claw member **75**. The claw member **75** includes a claw **750** which can engage with the engagement rotation member **720** (the protrusions and recesses in the outer circumferential surface thereof). The claw member **75** is supported to be turnable in a direction approaching the engagement rotation member **720** and in a direction separating from the engagement rotation member **720**.

(51) The claw member **75** is turned in the direction approaching the engagement rotation member **720** to engage with the engagement rotation member **720**. The claw **750** engages with any one of the protrusions **721**, and thus the claw member **75** engages with the engagement rotation member **720**. The claw member **75** is turned in the direction separating from the engagement rotation member **720**, and thus the engagement of the engagement rotation member **720** with the claw member **75** is released.

(52) The switching mechanism **7** also includes a solenoid **76**. The solenoid **76** includes a movable member **761** and a tension coil spring **762**. The movable member **761** is coupled to the claw member **75**. The position of the movable member **761** is displaced, and thus the claw member **75** approaches the engagement rotation member **720** or the claw member **75** separates from the engagement rotation member **720**. One end of the tension coil spring **762** is coupled to the movable member **761**.

(53) The solenoid **76** generates a magnetic force to displace the position of the movable member **761** against the tensile force of the tension coil spring **762**, and thereby moves the claw member **75** in the direction approaching the engagement rotation member **720**. In other words, the solenoid **76** generates the magnetic force to cause the claw member **75** to engage with the engagement rotation member **720**.

(54) On the other hand, when the claw member **75** is moved in the direction separating from the engagement rotation member **720**, the solenoid **76** stops the generation of the magnetic force. In this way, by the tensile force of the tension coil spring **762**, the claw member **75** is moved in the direction separating from the engagement rotation member **720**. In other words, the engagement of the engagement rotation member **720** with the claw member **75** is released.

(55) The claw member **75** engages with the engagement rotation member **720**, and thus the engagement rotation member **720** cannot be rotated, with the result that the sun gear **72** cannot be

rotated. In this way, the first gear **1** is rotated in the one direction (that is, in a direction which turns the push-up member **102** in the lift-up direction).

(56) On the other hand, the engagement of the engagement rotation member **720** with the claw member **75** is released, and thus the engagement rotation member **720** can be rotated, with the result that the sun gear **72** can be rotated. In this way, the first gear **1** is not rotated in the one direction. In other words, the power of the motor **M** is not transmitted to the first gear **1**. Here, the first gear **1** is brought into the state where the first gear **1** can be rotated in the other direction opposite to the one direction.

(57) <Restriction of Turning of Push-Up Member in Direction Opposite to Lift-Up Direction>

(58) The sheet storage device **100** includes a turning restriction member **8** as shown in FIGS. **5** and **6**. The turning restriction member **8** is made of resin. The turning restriction member **8** restricts the turning of the push-up member **102** in a direction opposite to the lift-up direction. The turning restriction member **8** is supported by the cylindrical member **10**. The turning restriction member **8** restricts the rotation of the second gear **2** when the transmission of the power from the motor **M** to the first gear **1** is blocked so as to restrict the turning of the push-up member **102** in the direction opposite to the lift-up direction.

(59) The turning restriction member **8** includes an annular portion **81**. The annular portion **81** is annular when viewed in the axial direction (that is, the width direction). The inner edge of the annular portion **81** is circular when the annular portion **81** is viewed in the axial direction. The outer shape of the annular portion **81** when viewed in the axial direction is not particularly limited. For example, the annular portion **81** is annular when viewed in the axial direction.

(60) The large diameter portion of the cylindrical member **10** is fitted to the annular portion **81**, and thus the turning restriction member **8** is supported by the cylindrical member **10**. In this configuration, the large diameter portion of the cylindrical member **10** serves as the support portion which supports the turning restriction member **8**. In the following description, the large diameter portion of the cylindrical member **10** is identified with a symbol **11**, and is referred to as the support portion **11** (see FIGS. **3** and **4**).

(61) The large diameter portion of the cylindrical member **10** is a part of the first gear **1**. In other words, the first gear **1** includes the cylindrical support portion **11** with its center axis being a cylindrical axis. The first gear **1** includes the support portion **11** on the one side in the axial direction.

(62) The turning restriction member **8** includes a protrusion portion **82** which protrudes outward in the radial direction from the annular portion **81**. The “outward in the radial direction” indicates a direction from the inside of the annular portion **81** (that is, the annular member) toward the outside. The protrusion portion **82** includes a plurality of protrusions which can engage with the output side gear **22** of the second gear **2**. The second gear **2** includes a cylindrical engagement portion which includes protrusions and recesses in the outer circumferential surface thereof with its center axis being a cylindrical axis. The output side gear **22** corresponds to the “engagement portion”.

(63) The turning restriction member **8** is rotatable around the axis of the first gear **1**. Specifically, the annular portion **81** is slidable on the outer circumferential surface of the support portion **11** around the axis of the first gear **1**. A lubricant such as grease may be arranged between the outer circumferential surface of the support portion **11** and the inner circumferential surface of the annular portion **81**.

(64) The annular portion **81** is spaced relative to the output side gear **22** (its tip end) in a direction orthogonal to the axial direction. In other words, when the output side gear **22** and the annular portion **81** are viewed in the axial direction, a gap **G** is provided therebetween. In this way, even when the annular portion **81** slides on the outer circumferential surface of the support portion **11** so as to rotate, the annular portion **81** does not interfere with the output side gear **22**.

(65) However, the protrusion portion **82** has a shape such that the protrusion portion **82** cannot pass through the gap **G** between the output side gear **22** and the annular portion **81** in the direction

orthogonal to the axial direction. In this way, although the annular portion **81** slides on the outer circumferential surface of the support portion **11**, and thus the turning restriction member **8** is rotated around the axis of the first gear **1**, when protrusion portion **82** reaches the gap G, the rotation of the turning restriction member **8** is stopped.

(66) Here, the protrusion portion **82** is located, relative to the gap G between the output side gear **22** and the annular portion **81**, on the downstream side in the direction of rotation of the output side gear **22** caused by the power of the motor M. In this way, the turning restriction member **8** has the function of restricting the turning of the push-up member **102** in the direction opposite to the lift-up direction.

(67) A specific description will be given below with reference to FIGS. 7 and 8. In FIGS. 7 and 8, the directions of the rotation of the second to fifth gears **2** to **5** are indicated by bold arrows. In FIGS. 7 and 8, the first gear **1** is not shown.

(68) When the power of the motor M is transmitted to the turning shaft **102a** of the push-up member **102** to turn the push-up member **102** in the lift-up direction, as shown in FIG. 7, the claw member **75** engages with the engagement rotation member **720**. Hence, the rotation of the engagement rotation member **720** is stopped.

(69) Here, the first gear **1** is rotated in the one direction (counterclockwise direction). The second gear **2** is rotated in the other direction (clockwise direction). The third to fifth gears **3** to **5** are respectively rotated in the directions indicated by the corresponding arrows. In this way, the power of the motor M is transmitted to the turning shaft **102a** of the push-up member **102**, and thus the push-up member **102** is turned in the lift-up direction. The lift plate **101** is raised.

(70) On the other hand, when the transmission of the power from the motor M to the turning shaft **102a** of the push-up member **102** is blocked, as shown in FIG. 8, the engagement of the engagement rotation member **720** with the claw member **75** is released. Hence, the engagement rotation member **720** can be rotated.

(71) Here, the rotation of the first gear **1** is stopped. Since the rotation of the first gear **1** is stopped, the rotation of the second gear **2** is also stopped. The rotation of the third to fifth gears **3** to **5** is also stopped. The engagement of the engagement rotation member **720** with the claw member **75** is released, and thus the first gear **1** can be rotated in the other direction.

(72) Here, when the engagement of the engagement rotation member **720** with the claw member **75** is released, the push-up member **102** attempts to be turned in the direction opposite to the lift-up direction by the weight of the lift plate **101** and the weight of the sheet S on the lift plate **101**. Here, the second to fifth gears **2** to **5** attempt to be rotated in the directions indicated by arrows in FIG. 8. The first gear **1** can be rotated in the other direction. Hence, when the second gear **2** attempts to be rotated in one direction (direction indicated by the arrow in FIG. 8), the first gear **1** attempts to be rotated in the other direction.

(73) If the second gear **2** is rotated in the one direction, the lift plate **101** is lowered. Consequently, the paper feed roller **1001** is prevented from making contact with the sheet S on the lift plate **101**, and thus a failure occurs in which the sheet S is not fed from the sheet storage device **100** to the conveyance path.

(74) Hence, in the present embodiment, the turning restriction member **8** is arranged in the support portion **11** of the first gear **1**. In this configuration, when the second gear **2** is rotated in the one direction (that is, when the second gear **2** is rotated in the direction opposite to the direction of rotation of the second gear **2** caused by the power of the motor M), the output side gear **22** engages with the protrusion portion **82** of the turning restriction member **8**, and in this state, the turning restriction member **8** is rotated around the axis of the first gear **1** relative to the outer circumferential surface of the cylindrical member **10**. Thereafter, although the protrusion portion **82** reaches the gap G between the output side gear **22** and the annular portion **81**, the protrusion portion **82** cannot pass through the gap G. In other words, the protrusion portion **82** reaches the gap G, and thus the rotation of the turning restriction member **8** is stopped. Here, the output side gear

22 engages with the protrusion portion **82**, and thus the second gear **2** is not rotated in the one direction any more. The second gear **2** is not rotated, and thus the third to fifth gears **3** to **5** are also not rotated.

(75) In this way, in the present embodiment, it is possible to suppress the turning of the push-up member **102** in the direction opposite to the lift-up direction. In other words, it is possible to suppress the unintentional lowering of the lift plate **101** on which the sheet S is placed.

Consequently, it is possible to suppress the occurrence of a failure in the feeding of the sheet S from the sheet storage device **100**.

(76) In the present embodiment, the turning restriction member **8** is formed with only the members shown in FIG. 5. In this way, it is possible to suppress the complication of the structure. For example, a worm wheel is arranged in a gear train, and thus it is possible to apply a brake to the first gear **1** using friction loss. However, when the worm wheel is arranged in the gear train, since the axial direction is changed, the number of components is increased, and the structure is complicated, with the result that this causes an increase in cost. Moreover, a drive loss when the push-up member **102** is turned in the lift-up direction is increased. On the other hand, the turning restriction member **8** is used, and thus it is possible to suppress the inconveniences described above.

(77) In the present embodiment, the annular portion **81** is fitted to the support portion **11**, and thus the turning restriction member **8** is supported. Here, the annular portion **81** is slidable on the outer circumferential surface of the support portion **11** in the direction of rotation thereof. In this configuration, when the first gear **1** is rotated in the one direction by the power of the motor M, by frictional resistance between the outer circumferential surface of the support portion **11** and the annular portion **81** and a force received from the second gear **2**, the annular portion **81** displaces a rotational position in a direction in which the protrusion portion **82** is moved away from the gap G. The force received by the annular portion **81** from the second gear **2** is a force with which the output side gear **22** turns the protrusion portion **82**. In this way, the rotation of the first gear **1** in the one direction is not prevented by the protrusion portion **82**.

(78) In the present embodiment, the output side gear **22** of the second gear **2** functions as the engagement portion which engages with the protrusion portion **82**. In this way, an engagement portion which engages with the turning restriction member **8** does not need to be provided separately. The shape of the engagement portion which engages with the protrusion portion **82** is not particularly limited, and it is sufficient that the engagement portion can engage with the protrusion portion **82**. In other words, the engagement portion does not need to be a gear.

(79) It should be considered that the embodiment disclosed herein is illustrative in all respects and not restrictive. The scope of the present disclosure is indicated not by the description of the above embodiment but by the scope of claims, and furthermore, meanings equivalent to the scope of claims and all changes within the scope are included therein.

Claims

1. A sheet storage device comprising: a lift plate on which a sheet is placed; a push-up member which is turnably supported below the lift plate, to which power of a motor is transmitted and which is turned in a lift-up direction to push up the lift plate; a first gear which is rotated in one direction by the power of the motor; a second gear which meshes with the first gear; a transmission mechanism which includes at least a third gear that meshes with the second gear to transmit the power of the motor to the push-up member; and a turning restriction member which restricts turning of the push-up member in a direction opposite to the lift-up direction, wherein the first gear includes a cylindrical support portion with a center axis of the first gear being a cylindrical axis, the turning restriction member includes an annular portion to which the support portion is fitted and which is slidable on an outer circumferential surface of the support portion in the direction of the

rotation and a protrusion portion which protrudes outward in a radial direction from the annular portion to be able to engage with the second gear and the protrusion portion has a shape such that the protrusion portion cannot pass through a gap between the second gear and the annular portion in a direction orthogonal to an axial direction and is located, relative to the gap, on a downstream side in a direction of rotation of the second gear caused by the power of the motor.

2. The sheet storage device according to claim 1, further comprising: a switching mechanism which switches between transmission of the power from the motor to the first gear and blocking of the transmission, wherein when the transmission of the power from the motor is blocked, the first gear can be rotated in another direction opposite to the one direction.

3. The sheet storage device according to claim 2, wherein when the transmission of the power from the motor to the first gear is blocked such that the second gear is rotated in a direction opposite to the direction of the rotation of the second gear caused by the power of the motor, the turning restriction member is rotated in a state where the protrusion portion engages with the second gear, and stops the rotation thereof when the protrusion portion reaches the gap.

4. The sheet storage device according to claim 1, wherein when the first gear is rotated in the one direction, by frictional resistance between the outer circumferential surface of the support portion and the annular portion and a force received from the second gear, the annular portion displaces a rotational position in a direction in which the protrusion portion is moved away from the gap.

5. The sheet storage device according to claim 1, wherein the second gear is a stepped gear and includes an input side gear which meshes with the first gear and an output side gear which meshes with the third gear, and the protrusion portion can engage with the output side gear.

6. An image forming apparatus comprising: the sheet storage device according to claim 1, wherein the image forming apparatus conveys the sheet and prints an image on the sheet.
